

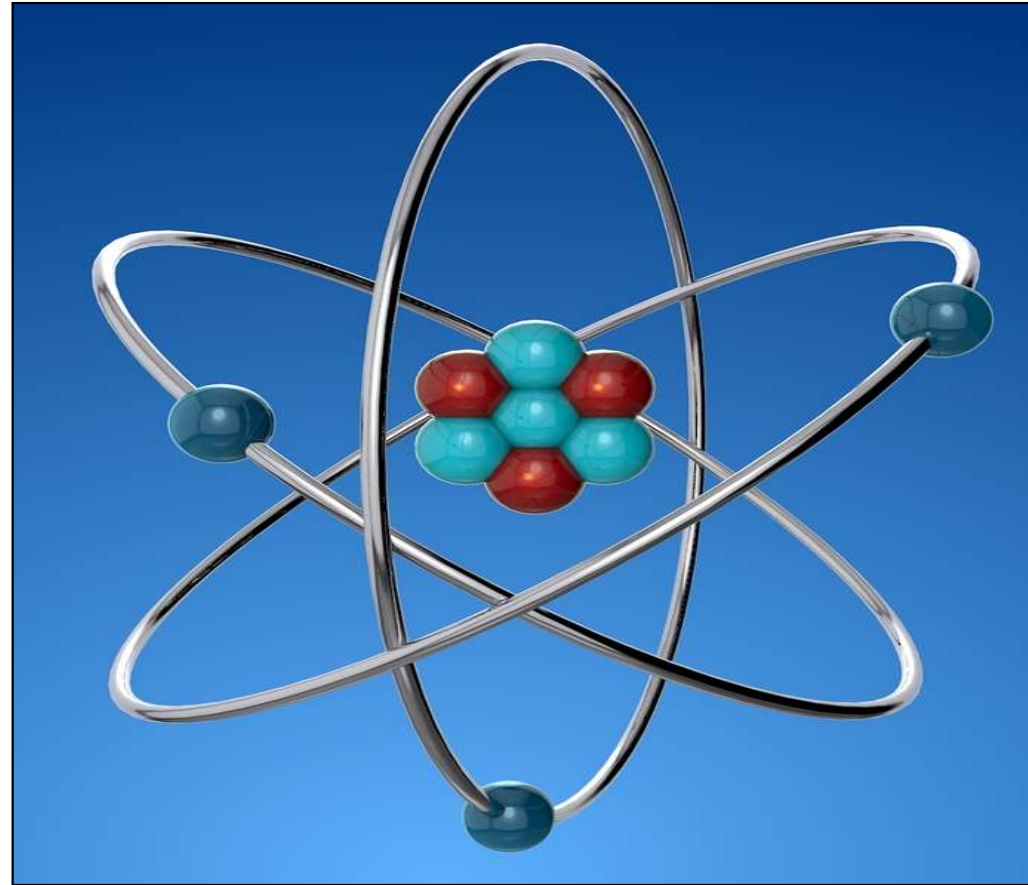
Lecture – 1 Introduction to Semiconductors and PN Junction Diodes



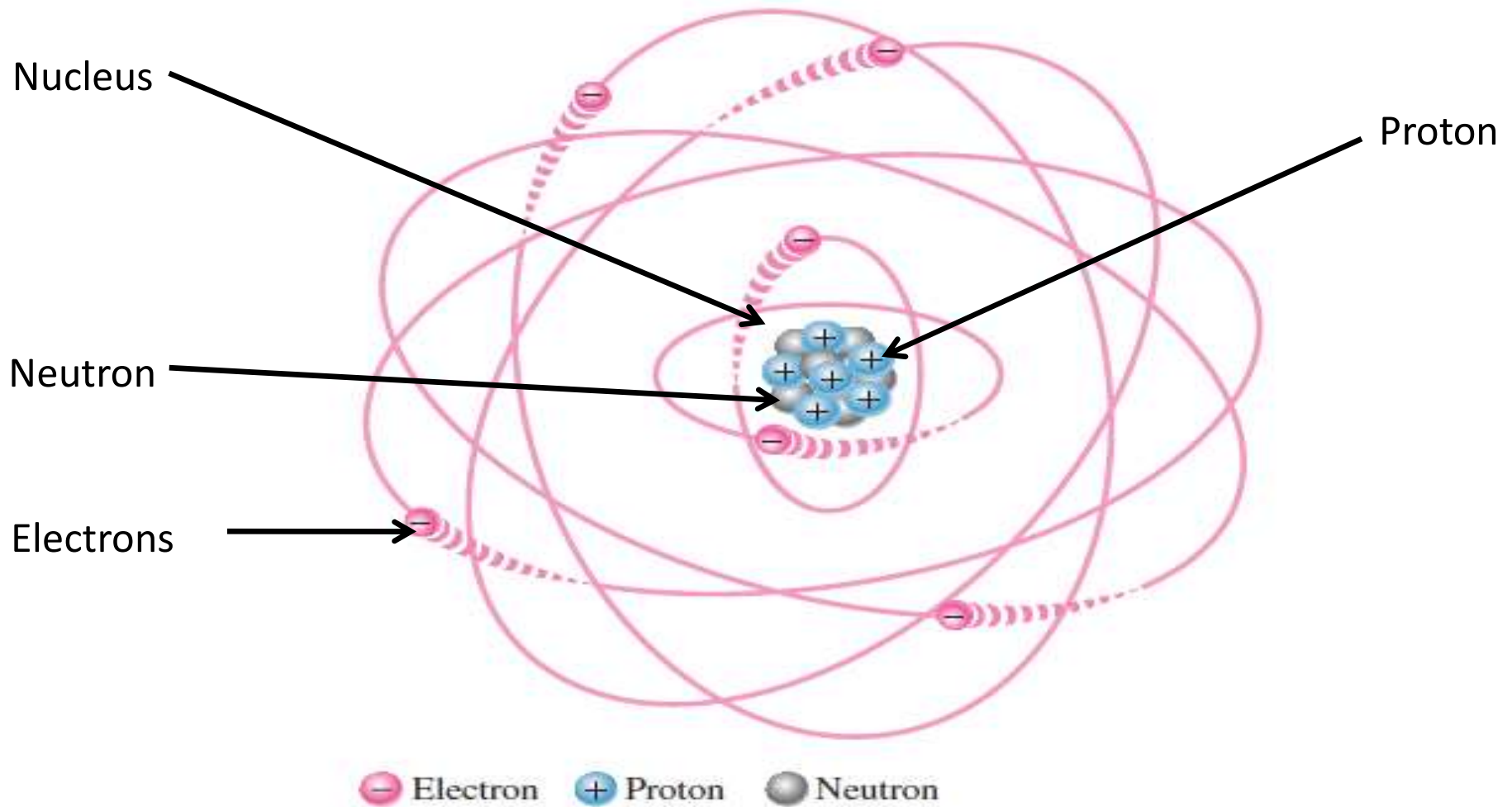
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The Atom – Bohr Model

- An atom* is the smallest particle of an element that retains the characteristics of that element.
- **Known 118** elements has atoms that are different from the atoms of all other elements.
- **Bohr model says atoms have a planetary type of structure** that consists of a central nucleus surrounded by orbiting electrons.

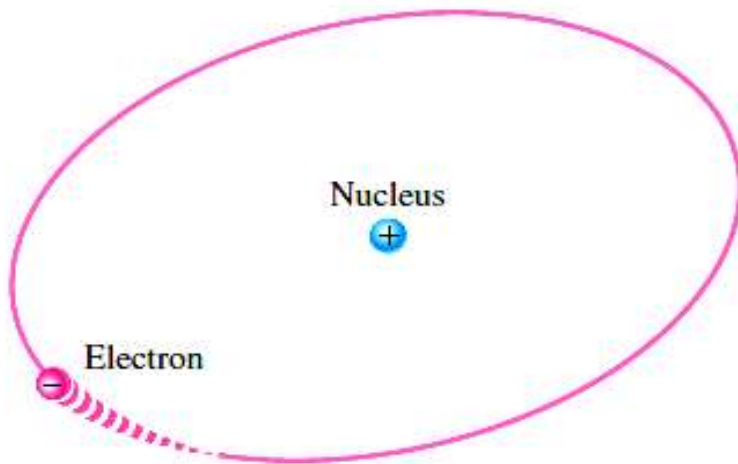


The Atom – Bohr Model

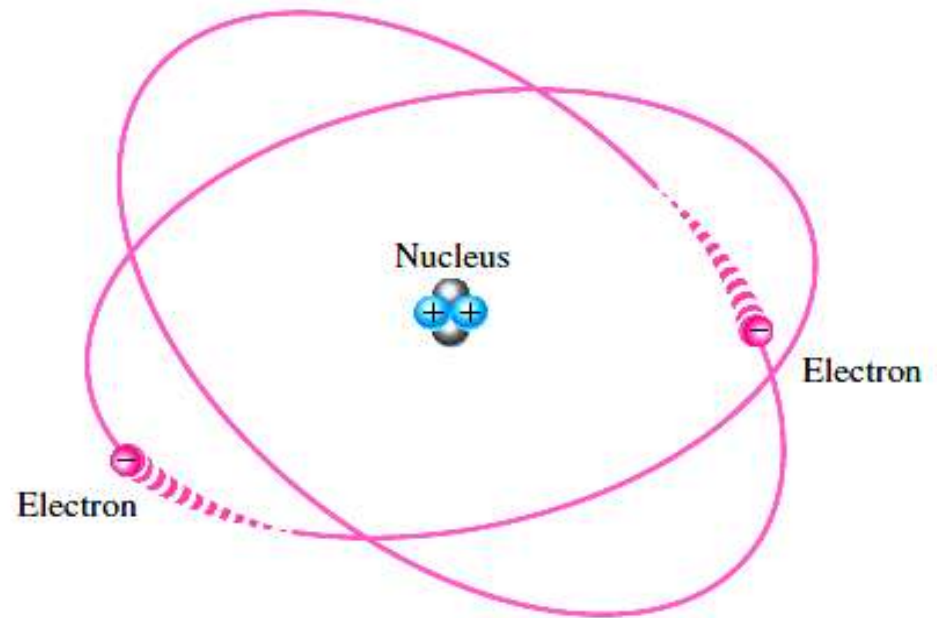


The Atom – Bohr Model

- Hydrogen, has one proton and one electron.



(a) Hydrogen atom



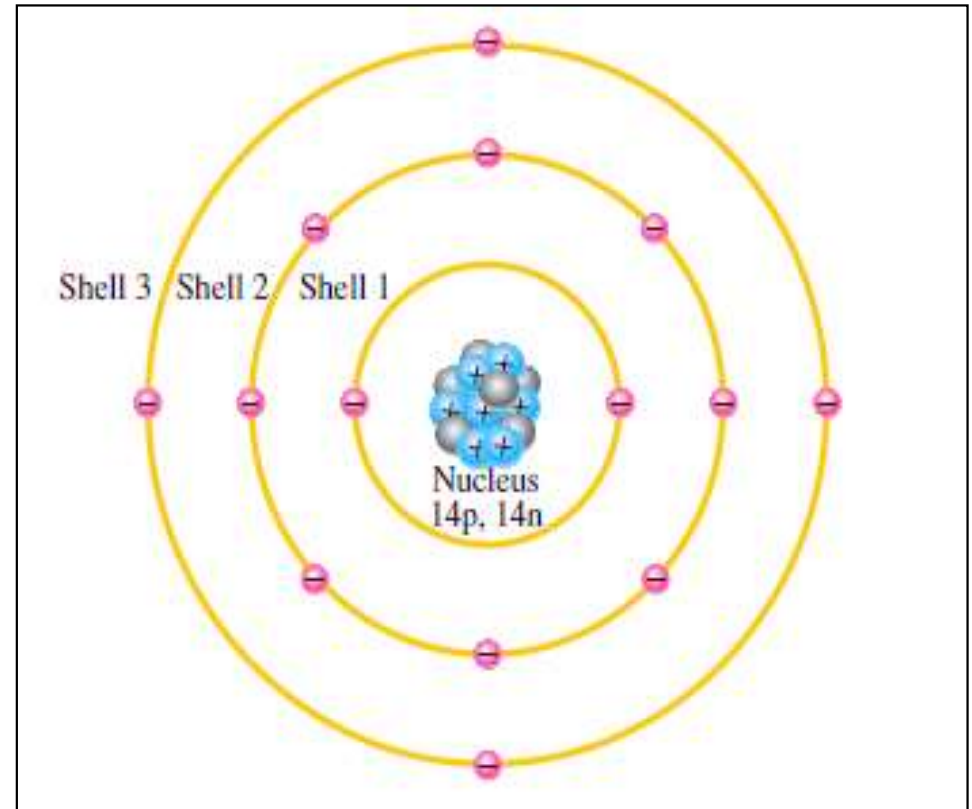
(b) Helium atom

- Helium atom, has two protons and two neutrons in the nucleus and two electrons orbiting the nucleus.

Electrons and Shells

Energy Levels

- *Electrons orbit the nucleus of an atom at certain distances from the nucleus.*
- Electrons **near the nucleus have less energy** than those in more distant orbits.
- Only **discrete (separate and distinct)** values of electron energies exist within atomic structures. Therefore, electrons must orbit only at discrete distances from the nucleus.
- **Each discrete distance (orbit) from the nucleus corresponds to a certain energy level.**
- **In an atom, the orbits are grouped into energy levels known as shells. A given atom has a fixed number of shells. Each shell has a fixed maximum number of electrons. The shells (energy levels) are designated 1, 2, 3, and so on, with 1 being closest to the nucleus.**



Electrons and Shells – Maximum

$$N_e = 2n^2$$

where n is the number of the shell. The maximum number of electrons that can exist in the innermost **shell (shell 1)** is;

$$N_e = 2n^2 = 2(1)^2 = 2$$

The maximum number of electrons that can exist in shell 2 is

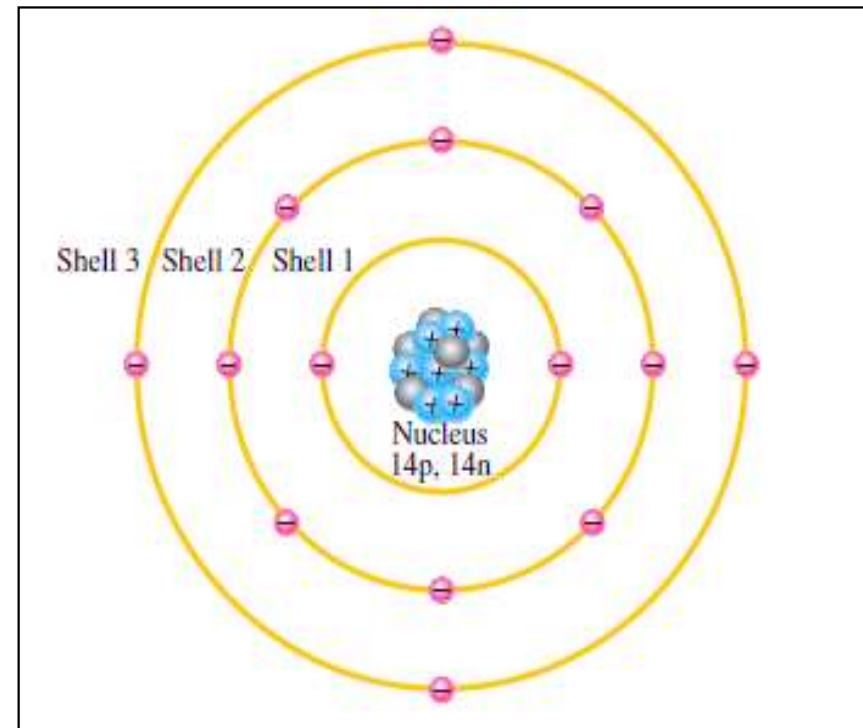
$$N_e = 2n^2 = 2(2)^2 = 2(4) = 8$$

The maximum number of electrons that can exist in shell 3 is

$$N_e = 2n^2 = 2(3)^2 = 2(9) = 18$$

The maximum number of electrons that can exist in shell 4 is

$$N_e = 2n^2 = 2(4)^2 = 2(16) = 32$$

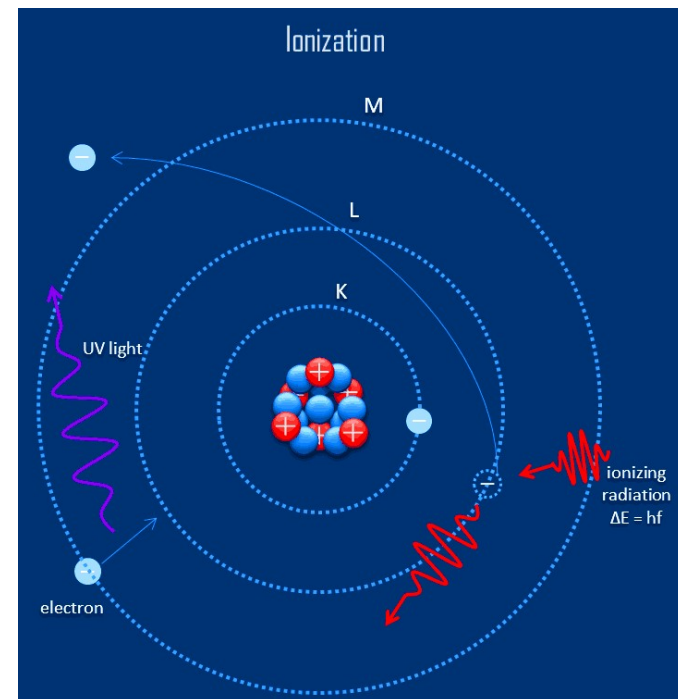


Valence Electrons

- Electrons that are in orbits farther from the nucleus have higher energy and are less tightly bound to the atom than those closer to the nucleus.
- This is because the force of attraction between the positively charged nucleus and the negatively charged electron decreases with increasing distance from the nucleus.
- Electrons with the highest energy exist in the outermost shell of an atom and are relatively loosely bound to the atom. This outermost shell is known as the **valence shell** and **electrons in this shell are called *valence electrons***.
- ***These*** valence electrons contribute to chemical reactions and bonding within the structure of a material and determine its electrical properties.
- When a valence electron gains sufficient energy from an external source, it can break free from its atom. This is the basis for conduction in materials.

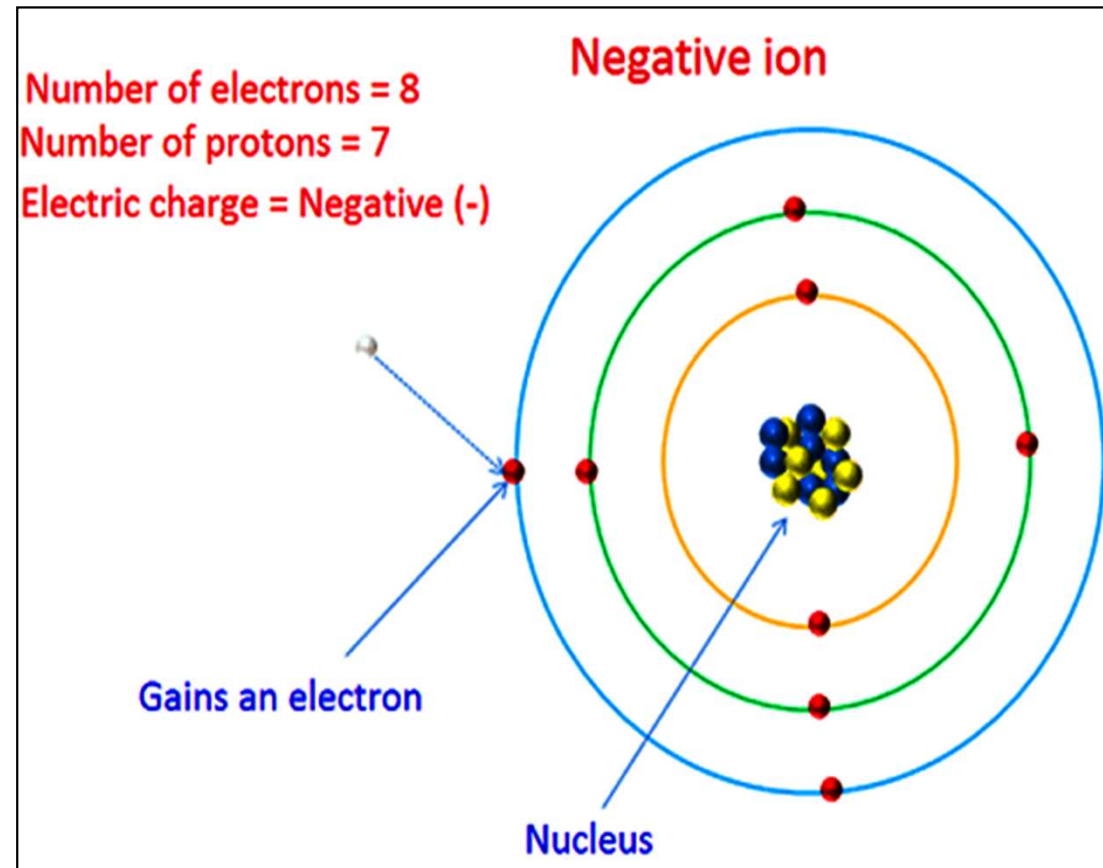
Ionization

- When an atom absorbs energy from a heat source or from light, the energies of the electrons are raised.
- The valence electrons possess more energy and are more loosely bound to the atom, jump to higher energy shells when external energy is absorbed by the atom.
- If a valence electron acquires a sufficient amount of energy, called **ionization energy**, it can actually escape from the outer shell and the atom's influence.
- Departure of a valence electron leaves a previously **neutral atom** with an excess of positive charge (more protons than electrons). The process of losing a valence electron is known as **ionization**, and the resulting positively charged atom is called a **positive ion**.
- When a neutral hydrogen atom loses its valence electron and becomes a positive ion, it is designated H. The escaped valence electron is called a **free electron**.



Ionization

- The reverse process can occur in certain atoms when a free electron collides with the atom and is captured, releasing energy. **The atom that has acquired the extra electron is called a *negative ion*.**
- ***The ionization process is not restricted to single atoms.*** In many chemical reactions, a group of atoms that are bonded together can lose or acquire one or more electrons.



Quantum Model

In the quantum model, each shell or energy level consists of up to four sub-shells called **orbital**, which are designated *s*, *p*, *d*, and *f*.

Orbital *s* = 2 *electrons*,
Orbital *p* = 6 *electrons*,
Orbital *d* = 10 *electrons*, and
Orbital *f* = 14 *electrons*.

NOTATION	EXPLANATION
$1s^2$	2 electrons in shell 1, orbital <i>s</i>
$2s^2 \quad 2p^3$	5 electrons in shell 2: 2 in orbital <i>s</i> , 3 in orbital <i>p</i>

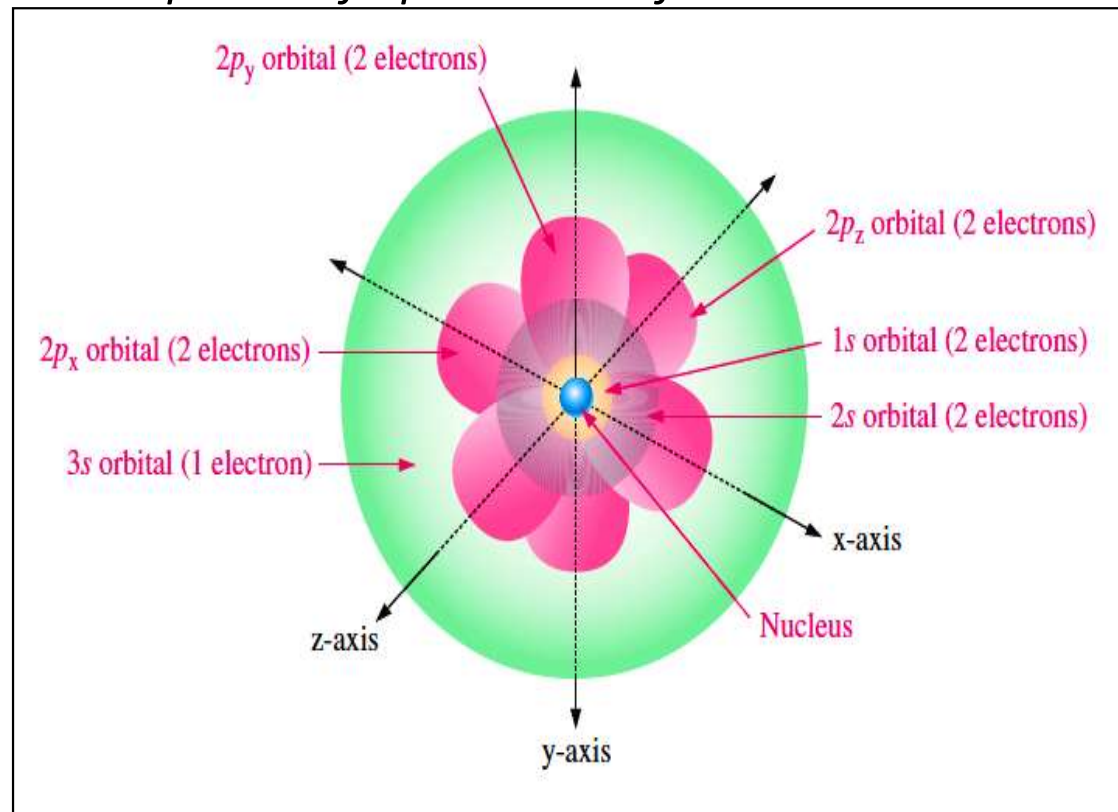
Each atom can be described by an electron configuration table that shows the shells or energy levels, the orbital, and the number of electrons in each orbital.

The first full-size number is the shell or energy level, the letter is the orbital, and the exponent is the number of electrons in the orbital.

Quantum Model

3D representation of the quantum model of an atom:

- ***s-orbitals*** are shaped like **spheres** and for energy level 1, the sphere is “solid” but for energy levels 2 or more, each single *s-orbital* is composed of spherical surfaces that are nested shells.
- A ***p-orbital*** for shell 2 has the form of two **ellipsoidal lobes** with a point of tangency at the nucleus (sometimes referred to as a **dumbbell shape**.)
- ***p-orbital*** in each energy level are oriented at right angles to each other. One is oriented on the x-axis, one on the y-axis, and one on the z-axis.

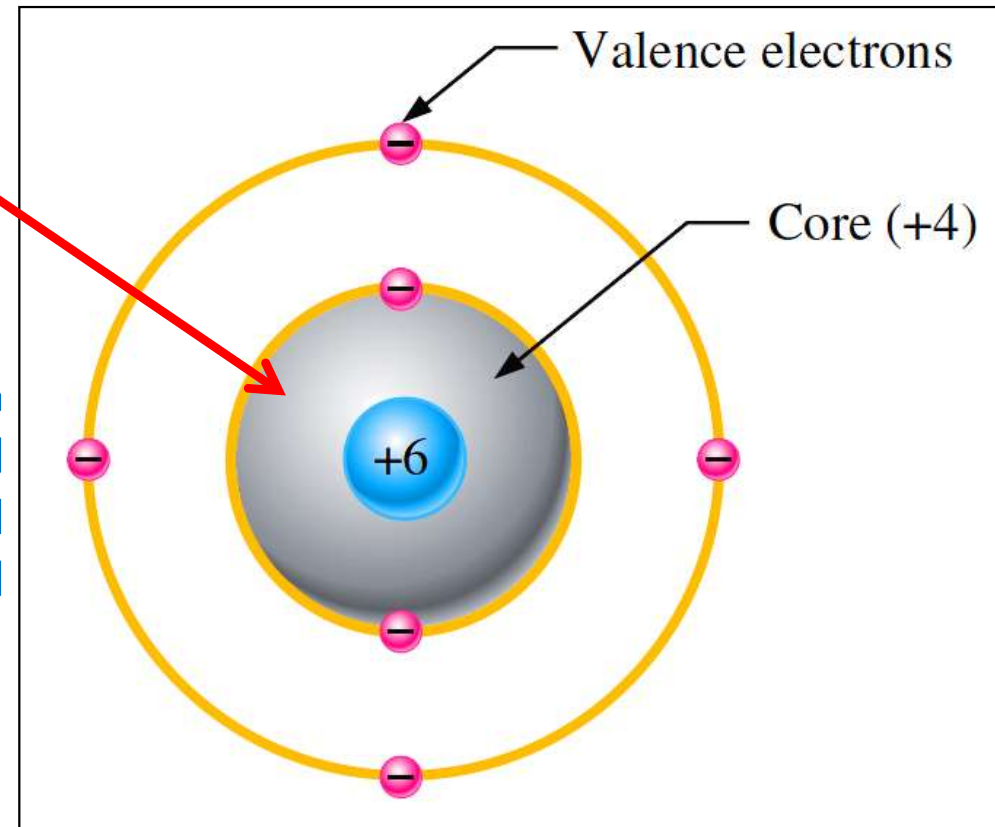


Insulators, Conductors, and Semiconductors

Electrical properties, an atom can be represented by the valence shell and a **core that consists of all the inner shells and the nucleus**.

$$\text{Core Charge} = +6 \text{ (proton)} - 2 \text{ (electron)} = +4$$

Understanding Valence shell and Core can be good for understanding the electrical conductivity.



Insulators, Conductors, and Semiconductors

Insulator

- An insulator is a material that does not conduct electrical current under normal conditions because of very high resistivities.
- Valence electrons are tightly bound to the atoms; therefore, there are very few free electrons in an insulator.
- **Examples** of insulators are rubber, plastics, glass, mica, and quartz.

Conductors

- ***A conductor is a material that easily conducts electrical current. Most*** metals are good conductors.
- The best conductors are single-element materials, such as copper (Cu), silver (Ag), gold (Au), and aluminum (Al), which are characterized by atoms with only one valence electron very loosely bound to the atom.
- These loosely bound valence electrons become free electrons. Therefore, in a conductive material the free electrons are valence electrons.

Insulators, Conductors, and Semiconductors

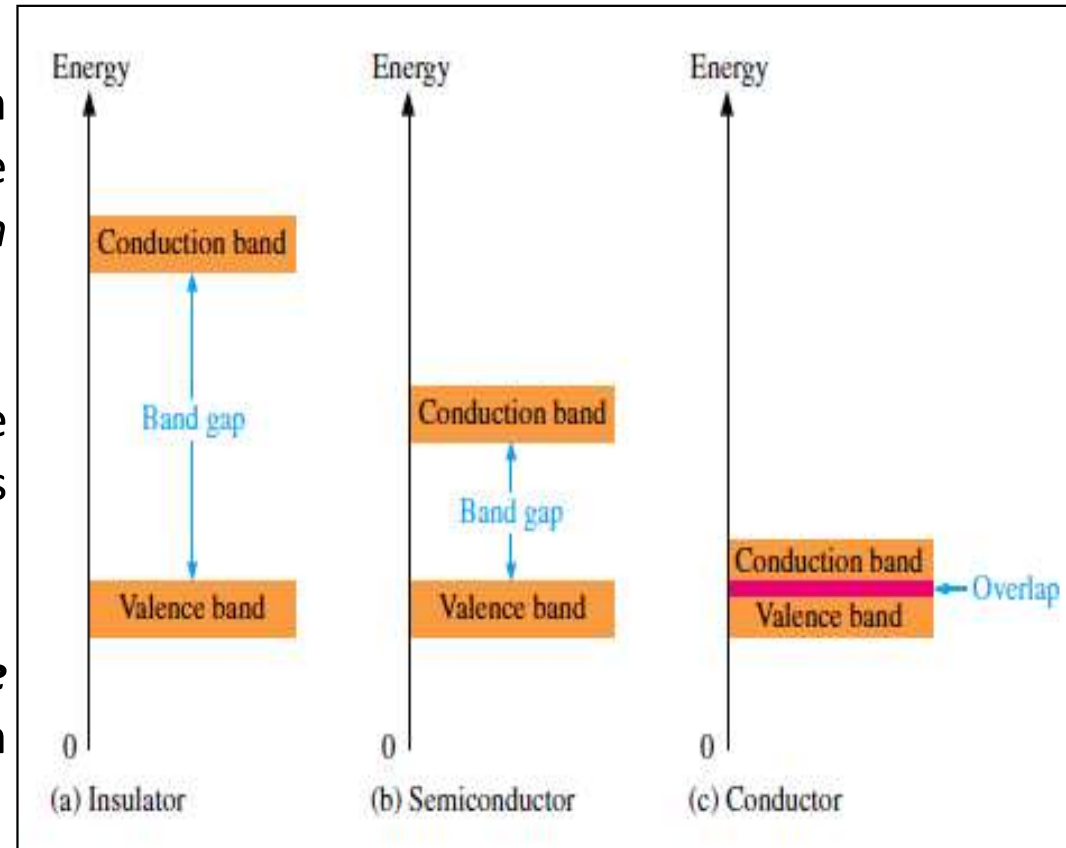
Semiconductor

- *A semiconductor is a material that is between conductors and insulators* in its ability to conduct electrical current.
- A semiconductor in its pure (intrinsic) state is neither a good conductor nor a good insulator. Single-element semiconductors are silicon (Si), and germanium (Ge).
- Antimony (Sb), Arsenic (As), Astatine (At), Boron (B), Polonium (Po), Tellurium (Te), are used as impurity atoms for doping Si or Ge crystals for increasing the utility or electrical property.
- Compound semiconductors (also called binary) such as Gallium arsenide, Indium Phosphide, Gallium nitride, Silicon carbide, and Silicon germanium are also commonly used.
- The single-element semiconductors are characterized by atoms with four valence electrons. **Silicon is the most commonly used semiconductor.**

Insulators, Conductors, and Semiconductors

Band Gap

- Valence shell of an atom represents a band of energy levels and that the valence electrons are confined to that band.
- When an electron acquires enough additional energy, it can leave the valence shell, become a *free electron*, and exist in what is known as the *conduction band*.
- The difference in energy between the valence band and the conduction band is called an *energy gap* or **band gap**.
- ***This is the amount of energy that a valence electron must*** have in order to jump from the valence band to the conduction band.
- Once in the conduction band, the electron is free to move throughout the material and is not tied to any given atom.

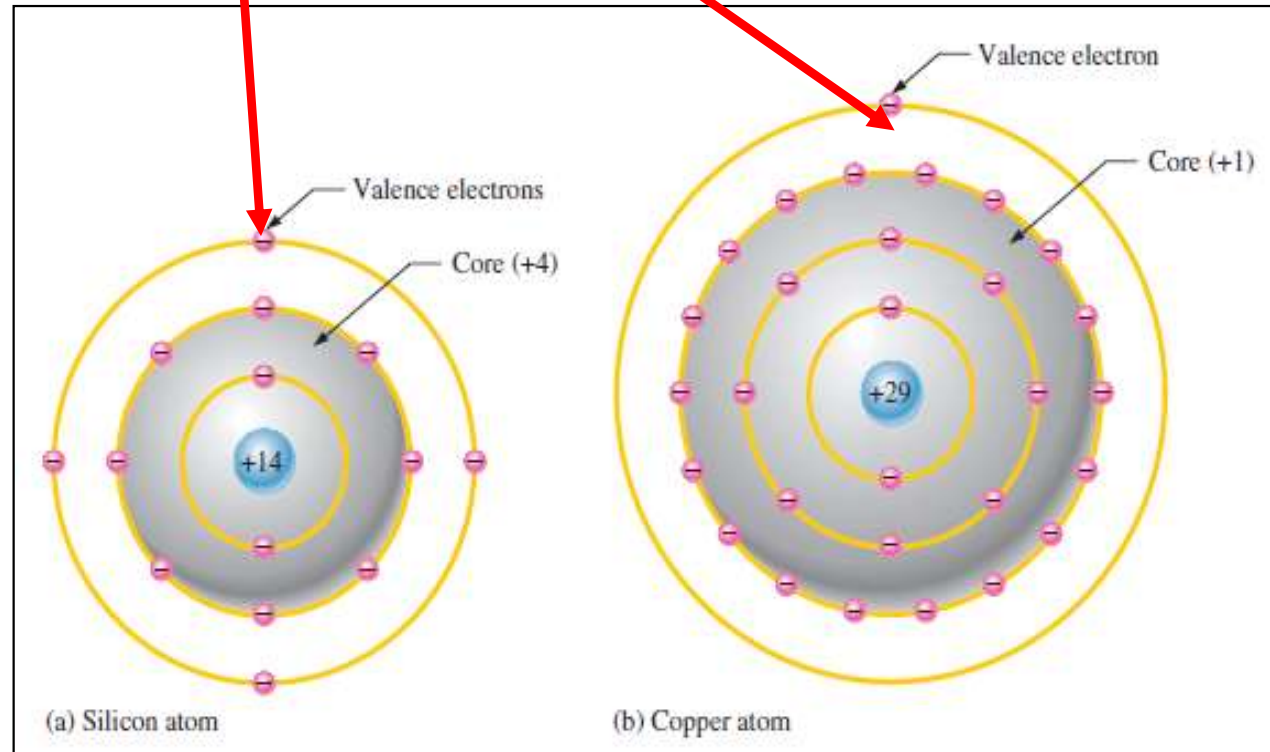


Comparison of a Semiconductor Atom to a Conductor Atom

$$\text{Core Charge} = +29 \text{ (proton)} - 28 \text{ (electron)} = +1$$

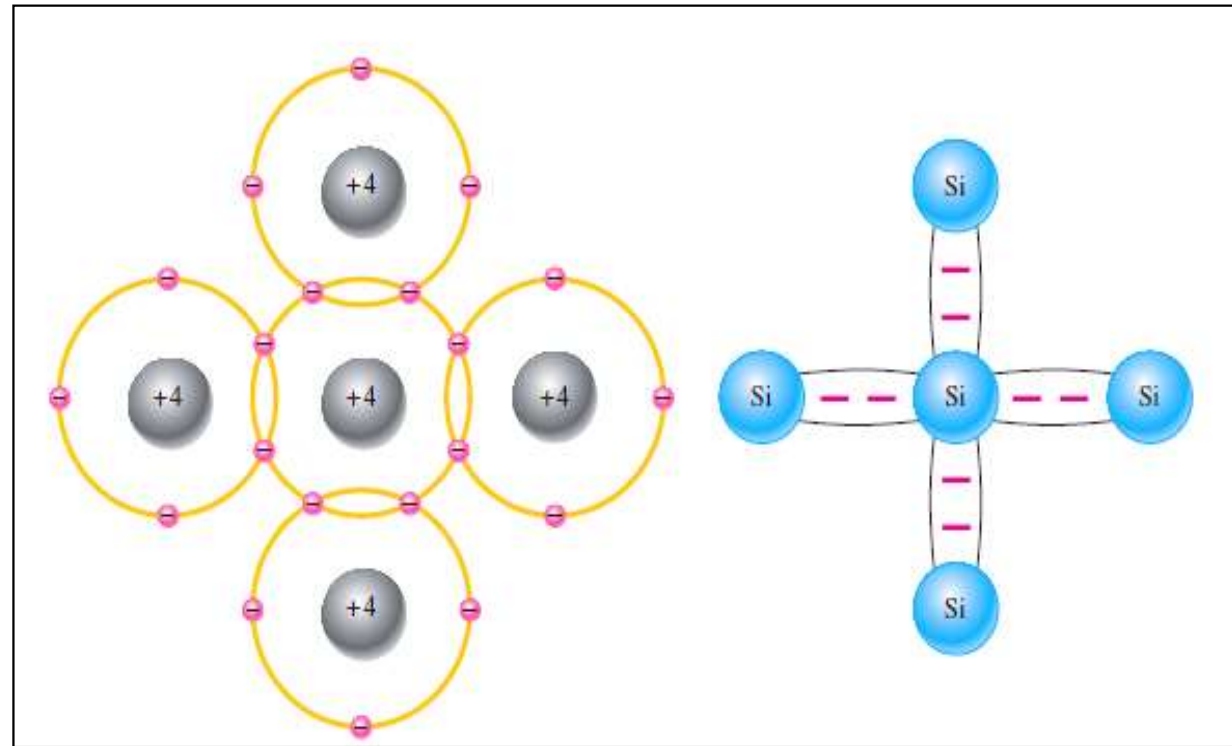
$$\text{Core Charge} = +14 \text{ (proton)} - 10 \text{ (electron)} = +4$$

- The valence electron in the copper atom “feels” an attractive force of 1.
- Valence electron in the silicon atom which “feels” an attractive force of 4.
- Therefore, there is more force trying to hold a valence electron to the atom in silicon than in copper.



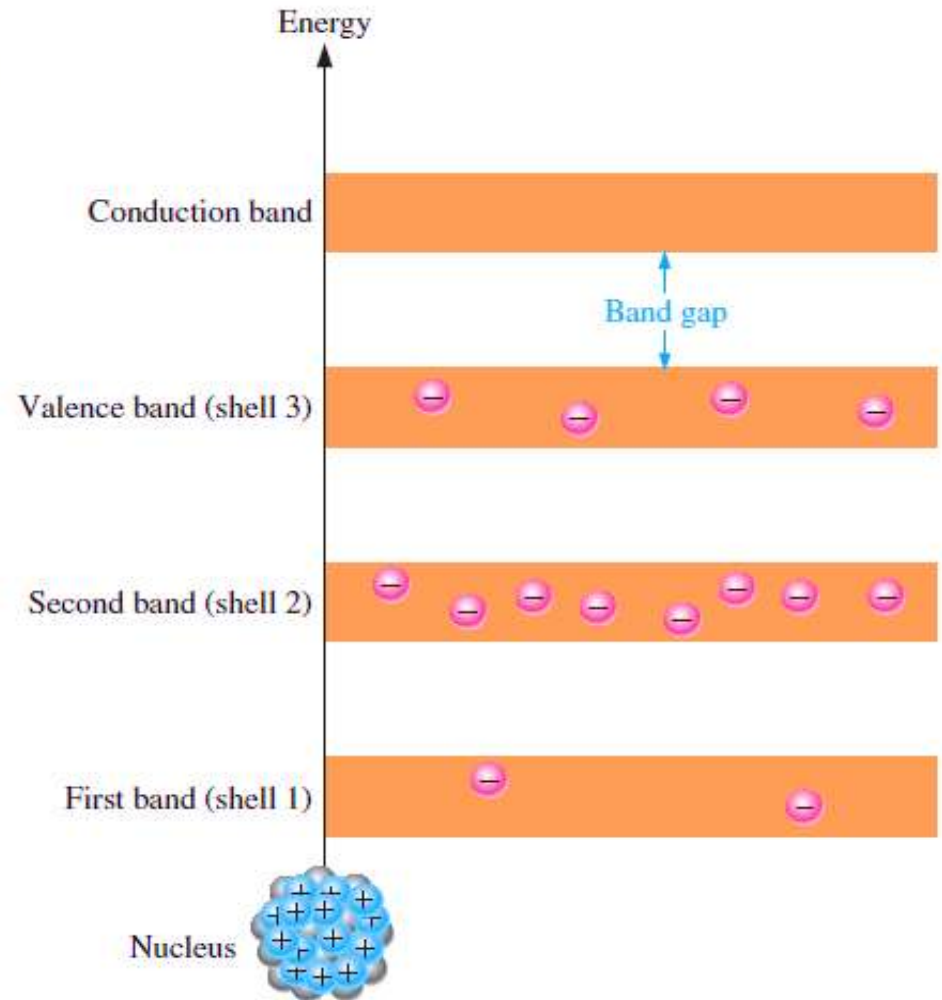
Covalent Bonds

- Silicon atom positions itself with four adjacent silicon atoms to form a silicon **crystal**.
- **A silicon (Si) atom with its four valence** electrons shares an electron with each of its four neighbors.
- This effectively creates eight shared valence electrons for each atom and produces a state of chemical stability.
- Sharing of valence electrons produces the **covalent bonds that hold the atoms together**; each valence electron is attracted equally by the two adjacent atoms which share it.



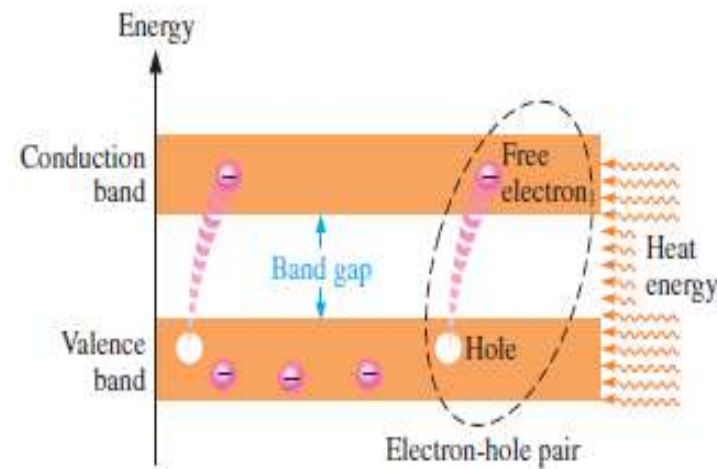
Current in Semiconductors – Unexcited

- At **0 K (-273 C)** the silicon crystal is very stable and all electrons are in its valence shell bonded with neighboring atom forming octet configuration.
- **Valence shell with 4 electrons and others as shown.**
- **They occupy the given energy level or the discrete energy level.**

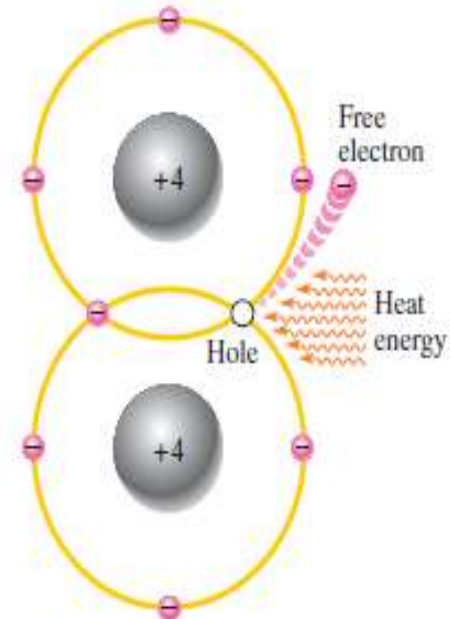


Conduction Electrons and Holes

- An **intrinsic (pure) silicon crystal** at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons.
- Free electrons are also called **conduction electrons**.



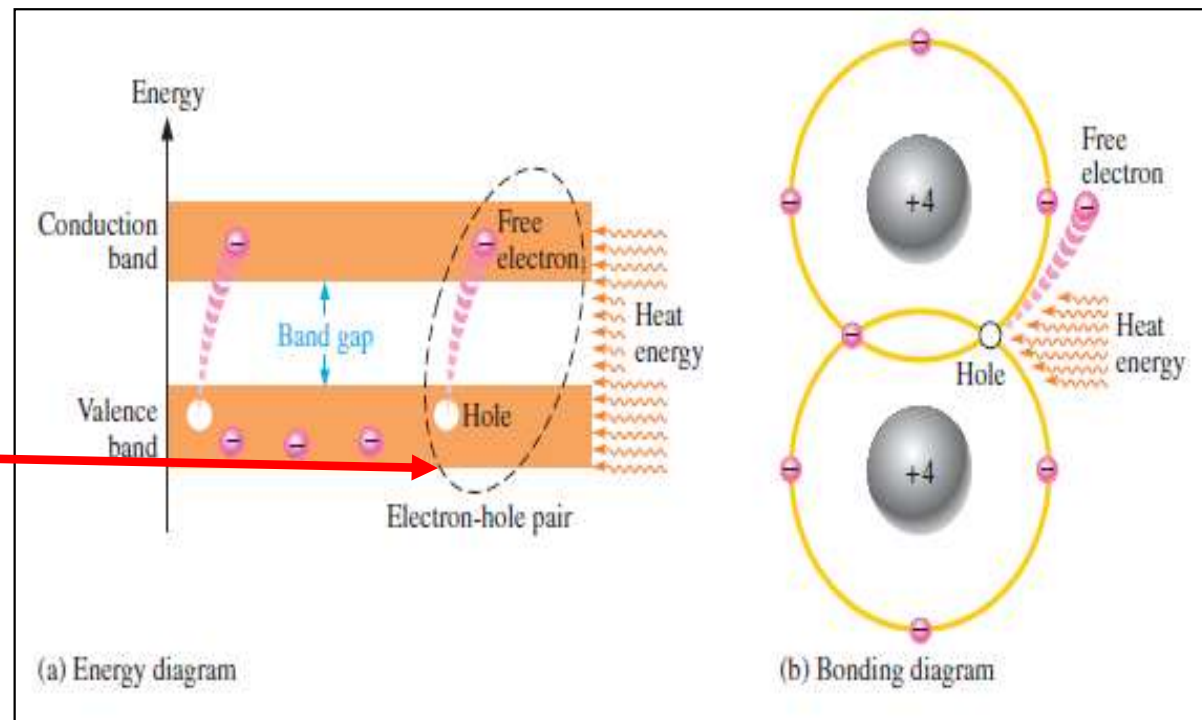
(a) Energy diagram



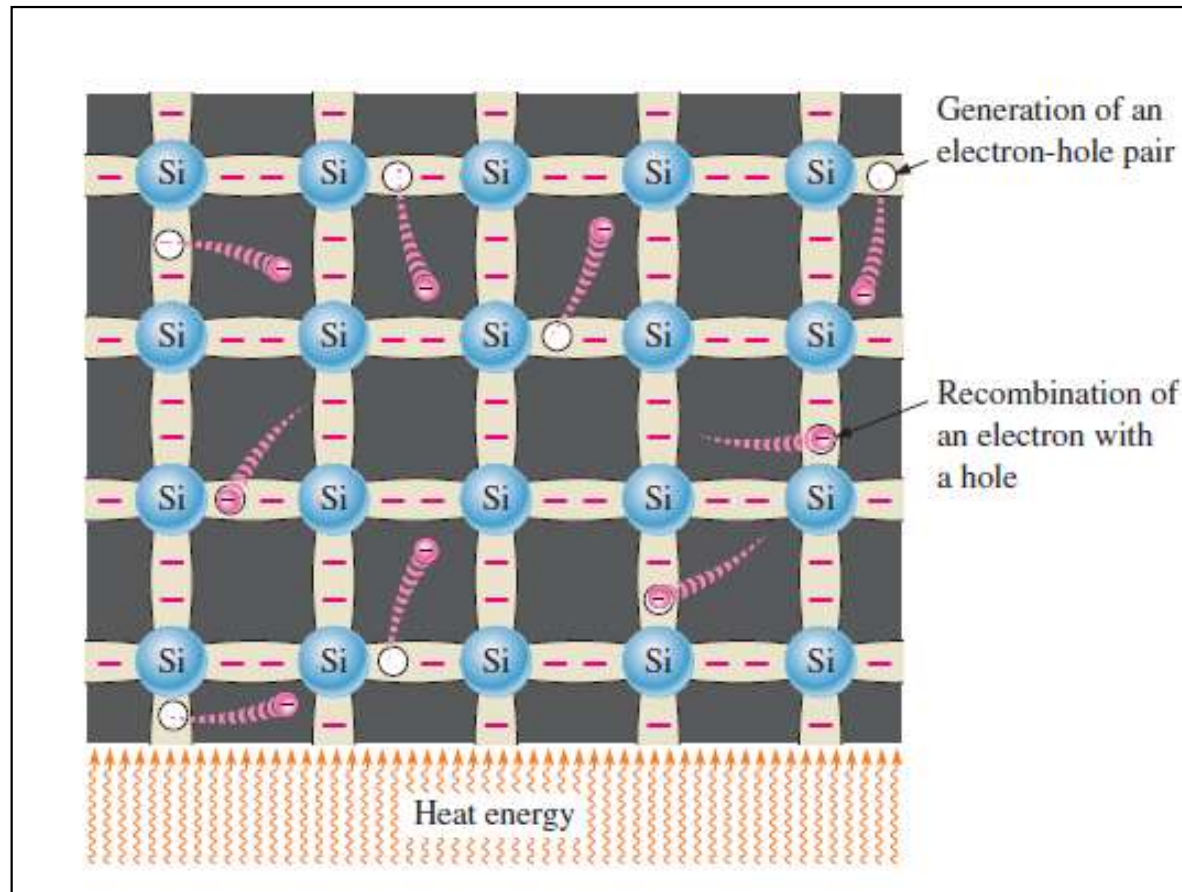
(b) Bonding diagram

Conduction Electrons and Holes

- An **intrinsic (pure) silicon crystal** at room temperature has sufficient heat (thermal) energy for some valence electrons to jump the gap from the valence band into the conduction band, becoming free electrons.
- Free electrons are also called **conduction electrons**.
- When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal.
- This vacancy is called a **hole**.
- **For every electron raised to the conduction band by external energy, there is one hole left in the valence band, creating what is called an electron-hole pair.**
- **Recombination** occurs when **conduction-band electron** loses energy and falls back into a hole in the valence band.

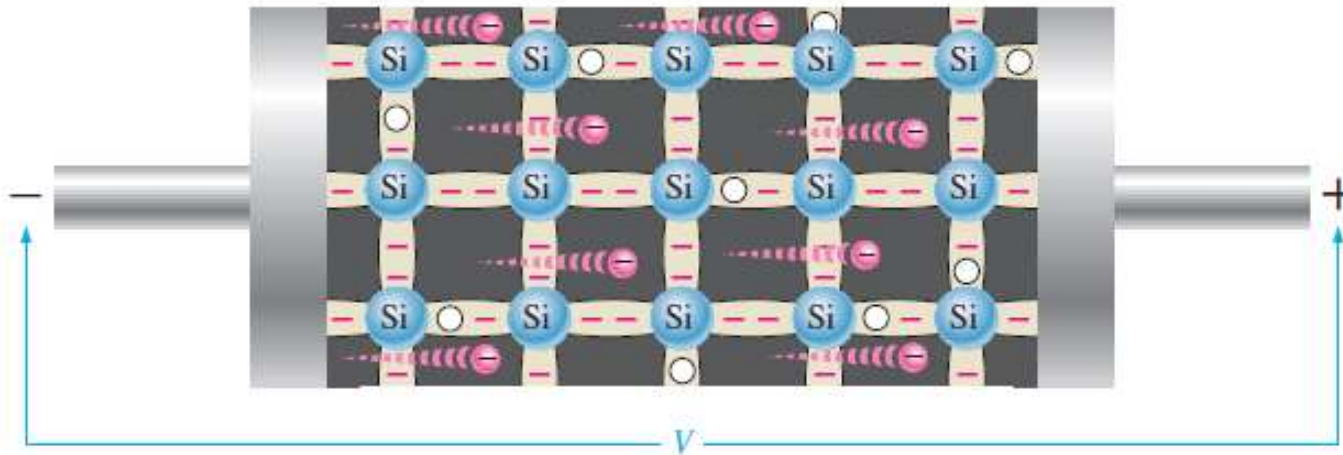


Conduction Electrons and Holes



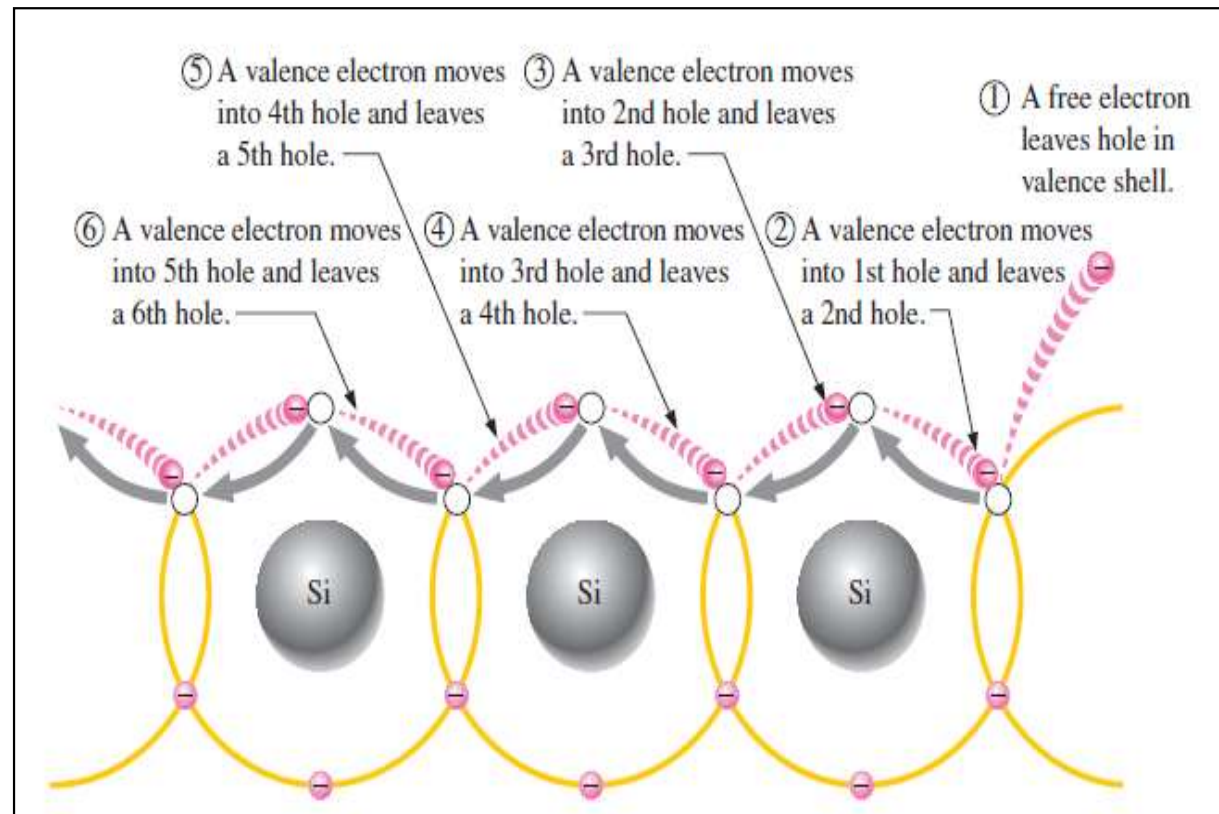
Electron and Hole Current

- When a voltage is applied across a piece of intrinsic silicon, as shown, the thermally generated free electrons in the conduction band, which are free to move randomly in the crystal structure, are now easily attracted toward the positive end.
- This movement of free electrons is one type of **current in a semi conductive material** and is called *electron current*.



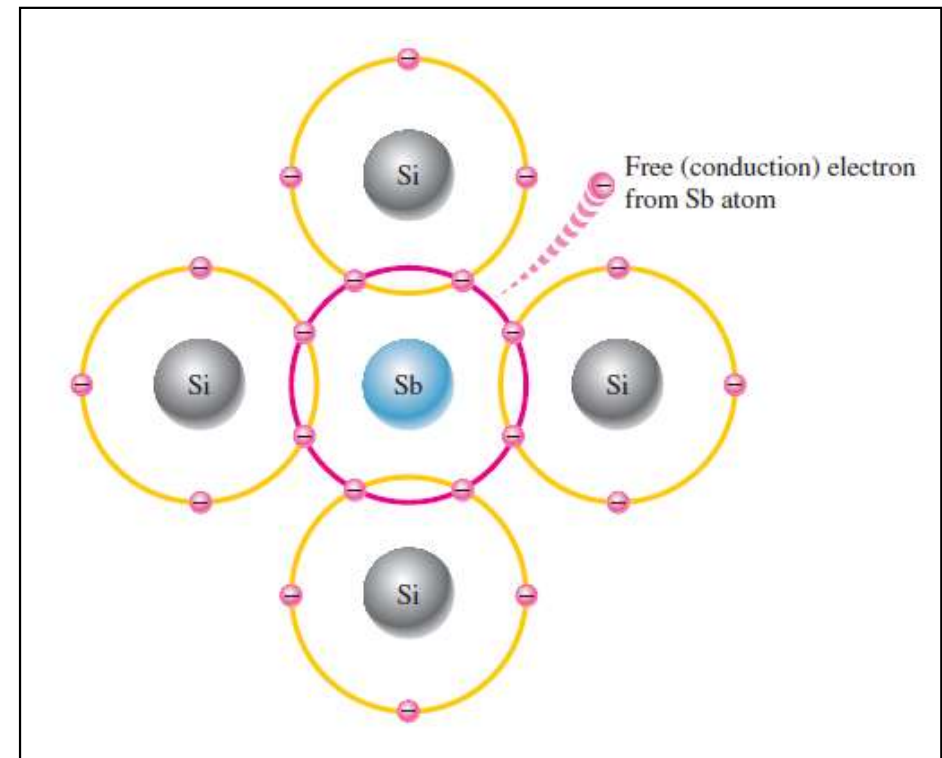
Hole current in intrinsic silicon.

- Another type of current occurs in the valence band, where the holes created by the free electrons exist.
- Electrons remaining in the valence band are still attached to their atoms and are not free to move randomly in the crystal structure as are the free electrons.
- However, a valence electron can move into a nearby hole with little change in its energy level, thus leaving another hole where it came from.
- Effectively the hole has moved from one place to another in the crystal structure.
- Although current in the valence band is produced by valence electrons, it is called *hole current* to distinguish it from electron current in the conduction band.



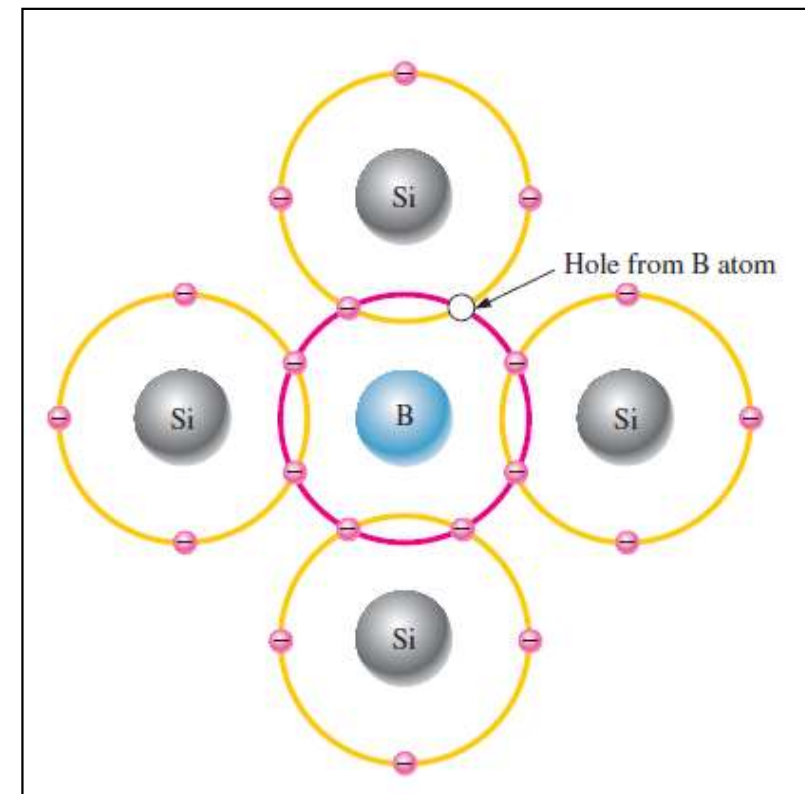
N-Type Semiconductor

- Pentavalent atom (antimony, in this case) forms covalent bonds with four adjacent silicon atoms. Four of the antimony atom's valence electrons are used to form the covalent bonds with silicon atoms, leaving one extra electron.
- This extra electron becomes a conduction electron because it is not involved in bonding.
- Because the Pentavalent atom gives up an electron, it is often called a *donor atom*.
- A conduction electron created by this doping process does not leave a hole in the valence band because it is in excess of the number required to fill the valence band.
- The electrons are called the **majority carriers in n-type material** but, there are also a few holes that are created when electron-hole pairs are thermally generated.
- These holes are *not produced by the addition of the pentavalent impurity atoms*. **Holes in an n-type material are called minority carriers.**

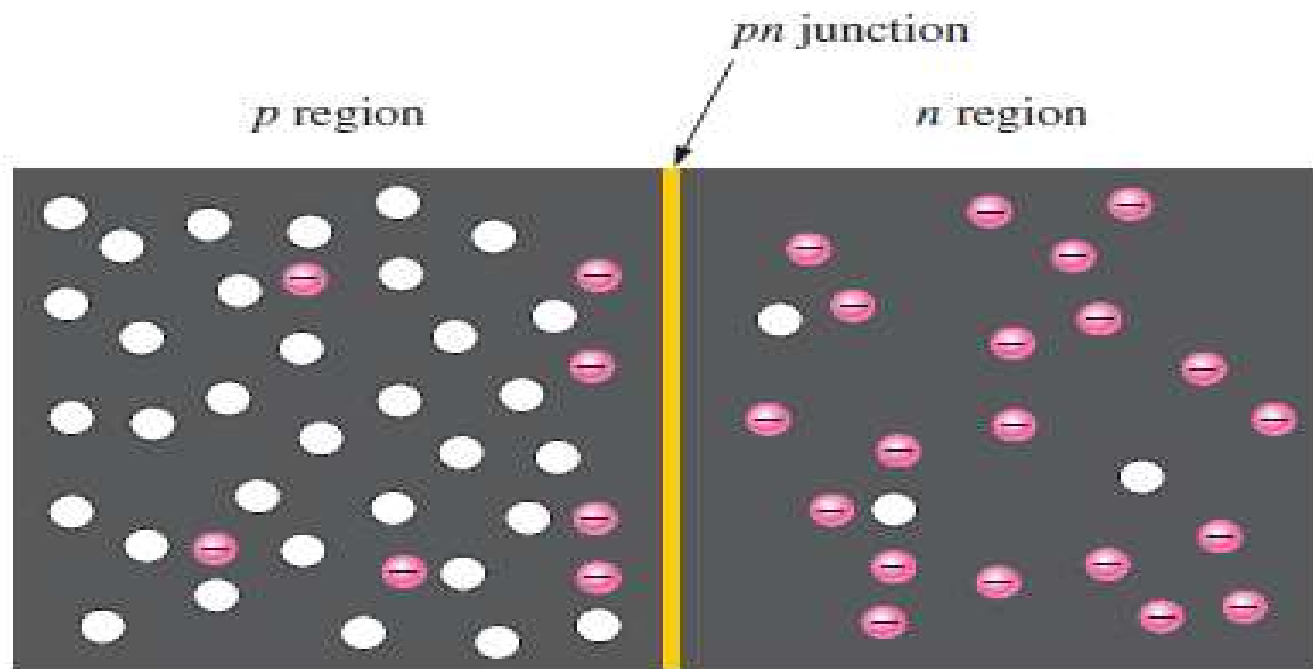


P-Type Semiconductor

- To increase the number of holes in intrinsic silicon, **trivalent impurity atoms are added**.
- These are atoms with three valence electrons such as boron (B), indium (In), and gallium (Ga).
- Each trivalent atom (Boron, in this case) forms covalent bonds with four adjacent silicon atoms.
- All three of the boron atom's valence electrons are used in the covalent bonds; and, since four electrons are required, a hole results when each trivalent atom is added.
- Because the trivalent atom can take an electron, it is often referred to as an *acceptor atom*.
- The holes are the majority carriers in *p-type material*. There are also a few conduction-band electrons that are created when electron-hole pairs are thermally generated.
- These conduction-band electrons are *not produced* by the addition of the trivalent impurity atoms and called minority carrier.

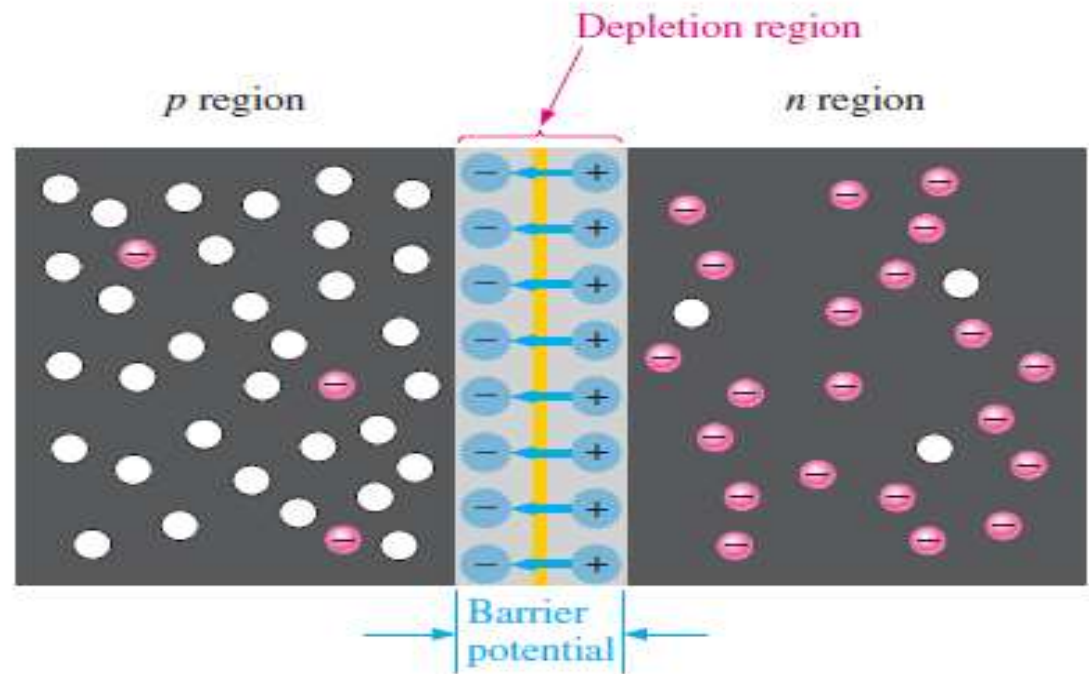


PN Junction Diode



PN Junction Diode

- When the *pn junction* is formed, the *n region* loses free electrons as they diffuse across the junction. This creates a layer of positive charges (pentavalent ions) near the junction.
- As the electrons move across the junction, the *p region* loses holes as the electrons and holes combine. This creates a layer of negative charges (trivalent ions) near the junction.
- These two layers of positive and negative charges form the **depletion region**.
- After the initial surge of free electrons across the *pn junction*, the *depletion region* has expanded to a point **where equilibrium is established** and there is no further diffusion of electrons across the junction.



THE END